

Scalability & Future Plan

Date	2 july 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited dataset size	May reduce prediction accuracy for real-world global data	High
2	Basic machine learning model (Linear Regression)	Lower accuracy compared to advanced models	High
3	No real-time external data integration	System depends only on static dataset	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users on Render free tier	Supports limited users on Render free tier
2	Data Storage	CSV-based local storage	Move to database system like MySQL or MongoDB
3	Performance	Basic Flask execution	Optimize model using advanced ML algorithms like Random Forest or XGBoost
4	Security	Simple login system	Implement JWT authentication and encrypted password storage

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Future Roadmap:

Phase 2	Improve ML model accuracy using advanced algorithms	Short-term	Better prediction accuracy
Phase 3	Add interactive dashboard with data visualization	Mid-term	Better user understanding of HDI trends
Phase 4	Integrate real-time global HDI dataset APIs	Long-term	More accurate and dynamic predictions